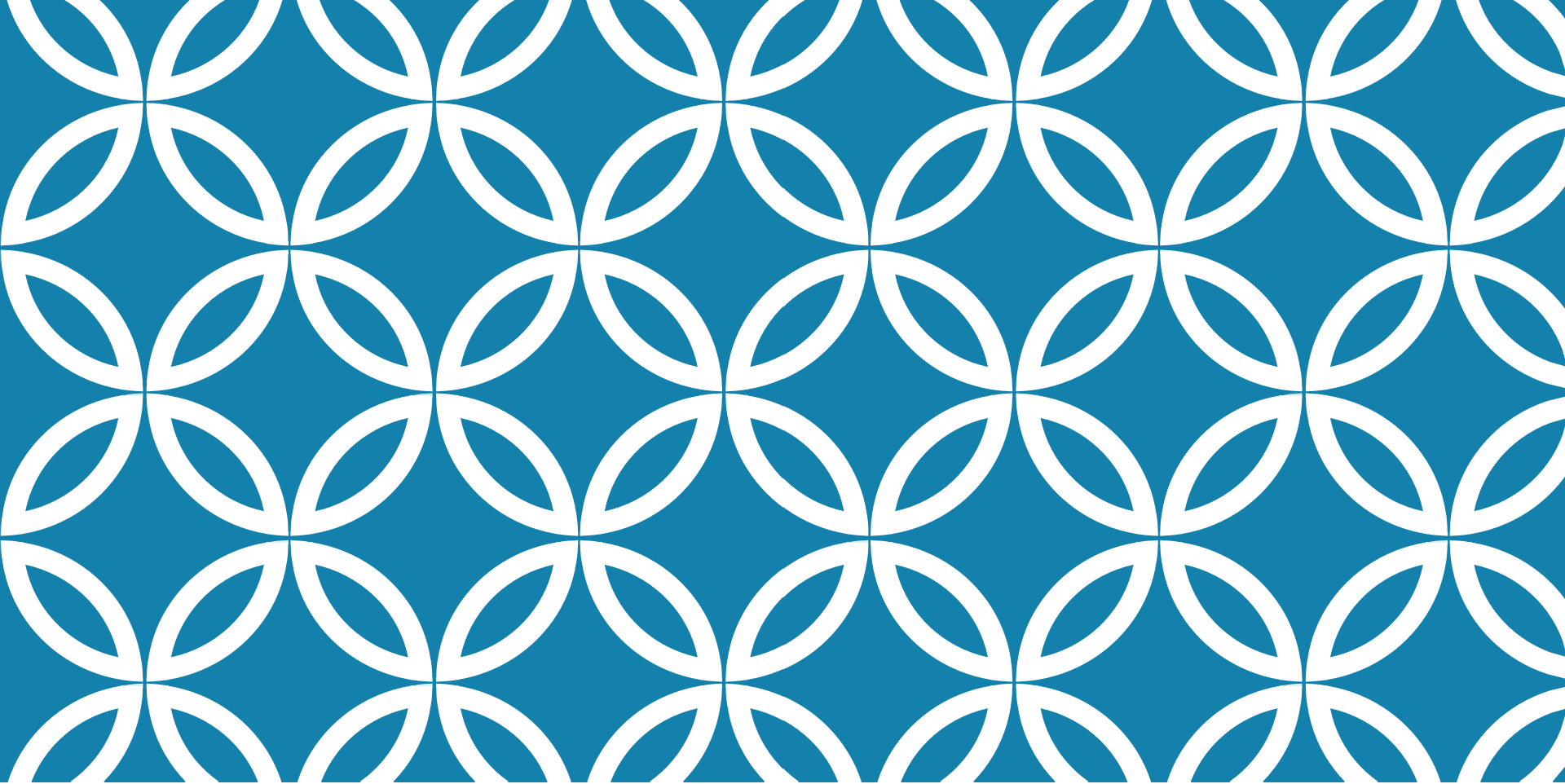




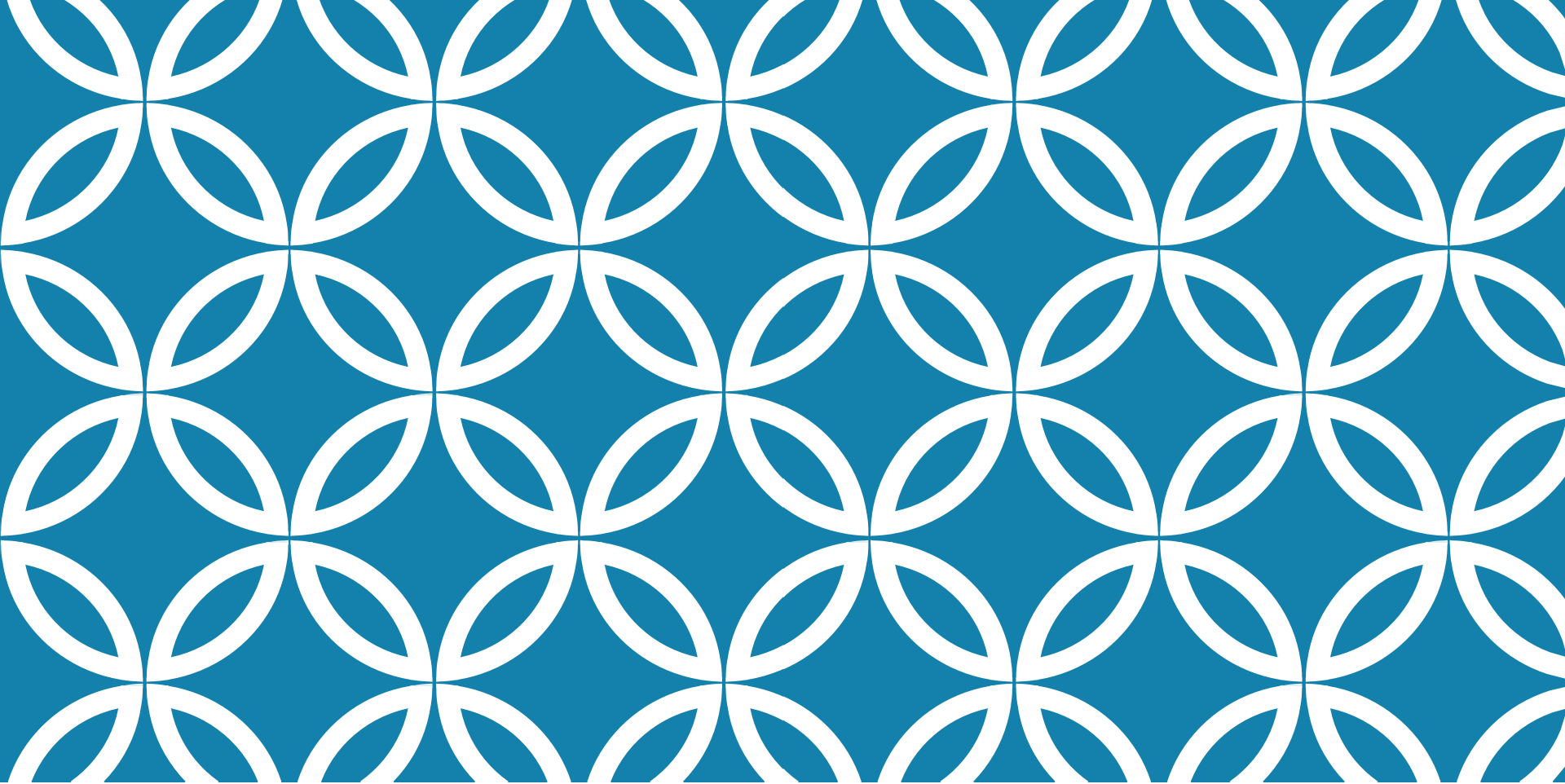
PROFESSIONAL ETHICS AND WORKING SAFETY

Lecturer: Dr
Ayesha

INTRODUCTION TO THE SUBJECT

Reference:

1. Quinn, M.J. 2011. Ethics for the information age. 2nd ed., Boston: Pearson Education, Inc.
2. George Reynolds, Ethics in Information Technology, Thomson Course Technology, Second Edition, 2006



CHAPTER 1

Introduction to
Ethics in
Computing

INTRODUCTION

Milestone in Computing, Networking & Information storage and retrieval

Impact of Information technology on society and social

Impact of information technology (IT) and the issues related

Ethics, Computer and privacy

INFORMATION AGE

Characterized by unprecedented (unique) access to information

Catalysts

- Low-cost computers
- High-speed communication networks
- Examples of advances in past two decades
 - Cell phones Email
 - World Wide Web
 - Cloud (Cloud computing is a way to access computing resources like storage, servers, and databases over the internet.)

TECHNOLOGY AND VALUES

Dynamic between people, technology

- ▢ People adopt technology
- ▢ Technology changes society

Different ways people are affected by technology

- ▢ Physical changes (e.g., laptops)
- ▢ Psychological changes (e.g., cell phones)

Technologies can solve problems, create new problems

- ▢ Automobile
- ▢ Refrigerator
- ▢ Low-cost international communication

CONTROL OVER NEW TECHNOLOGIES

People can control whether to adopt new technology

- ▢ Nuclear power

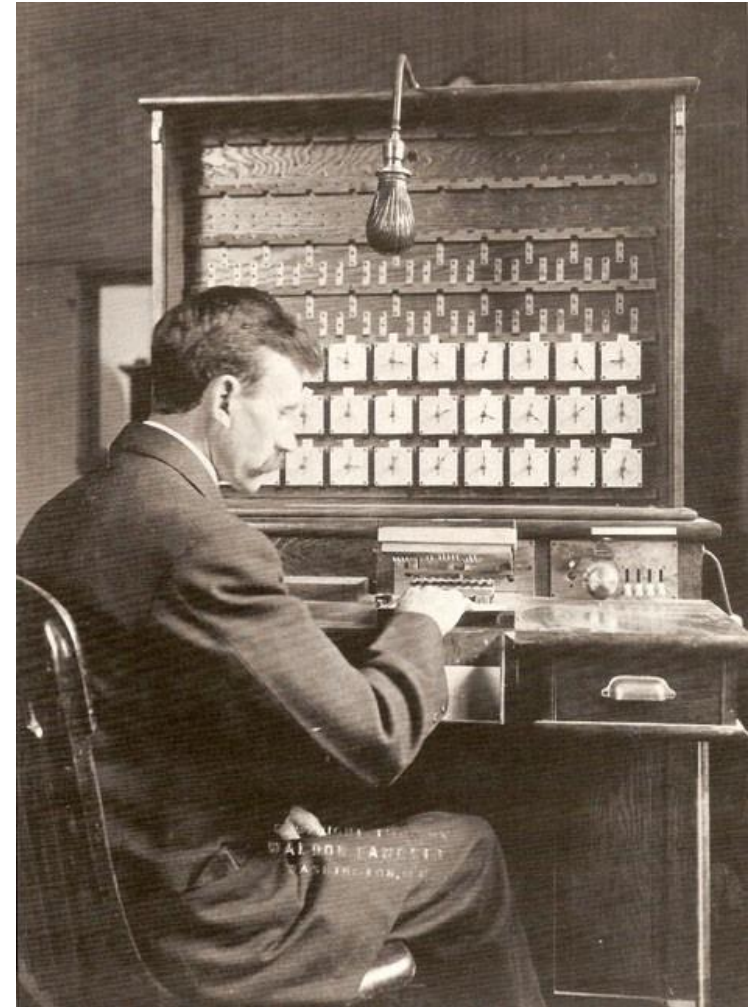
- ▢ People can influence rate at which technologies are developed

Intellectual property laws

- ▢ Tax structure

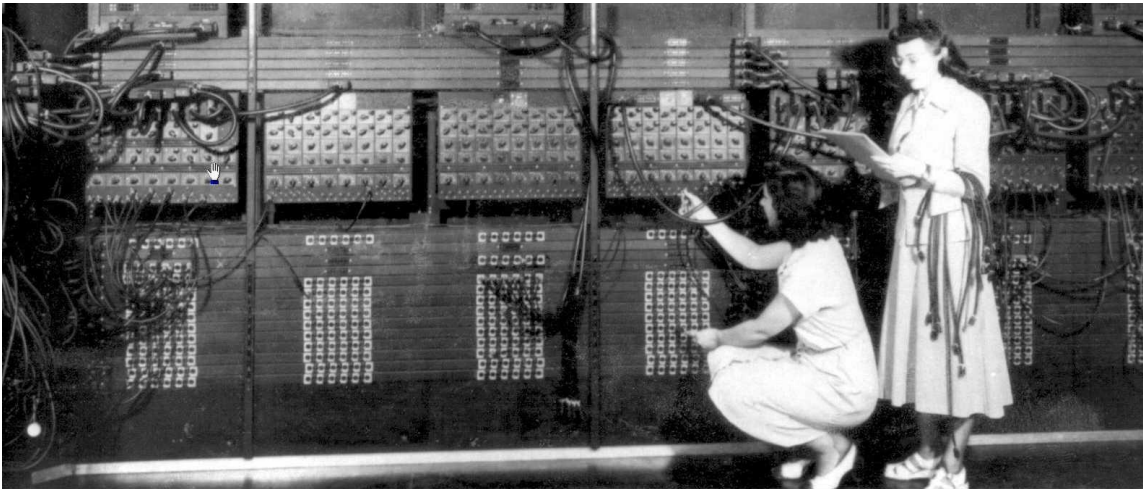
- ▢

MILESTONE IN COMPUTING



PUNCH CARD
TABULATOR

MILESTONE IN COMPUTING-CONTINUE



ENIAC

UNIVAC



MILESTONE IN COMPUTING-CONTINUE

AIDS TO MANUAL COMPUTING

-Abacus
(2000
years
ago)

Mechanical
Calculator
(1640)-
Blaise
Pascal

Cash
Register(
1879)-
James
Ritty

Punch
Card
Tabulation
(early
1800)-
Herman
Hollerith

Precursors of
Commercial
Computers-
Eniac(1946)

First
Commercial
Computers
(Univac,
1950)

Programming
Language and
Time-Sharing
(1957)-
Fortran,
(1959)-
COBOL

Transistor and
Integrated
Circuit
(1960-
1965)-
The
minuteman II
program

IBM
System/
360
(1964)

MILESTONE IN COMPUTING-CONTINUE

Microprocessor

-(1968)-Robert
Noyce and
Gordon Moore-
>Intel

-(1969)-Ted Hoff, Stanley
Mazor & Federico
Faggin->the first
microprocessor (Intel
4004)

Personal computer(1960- 1970)

(1975)Altair 8800-
>Altair Basic-
Microsoft (Bill
Gates)

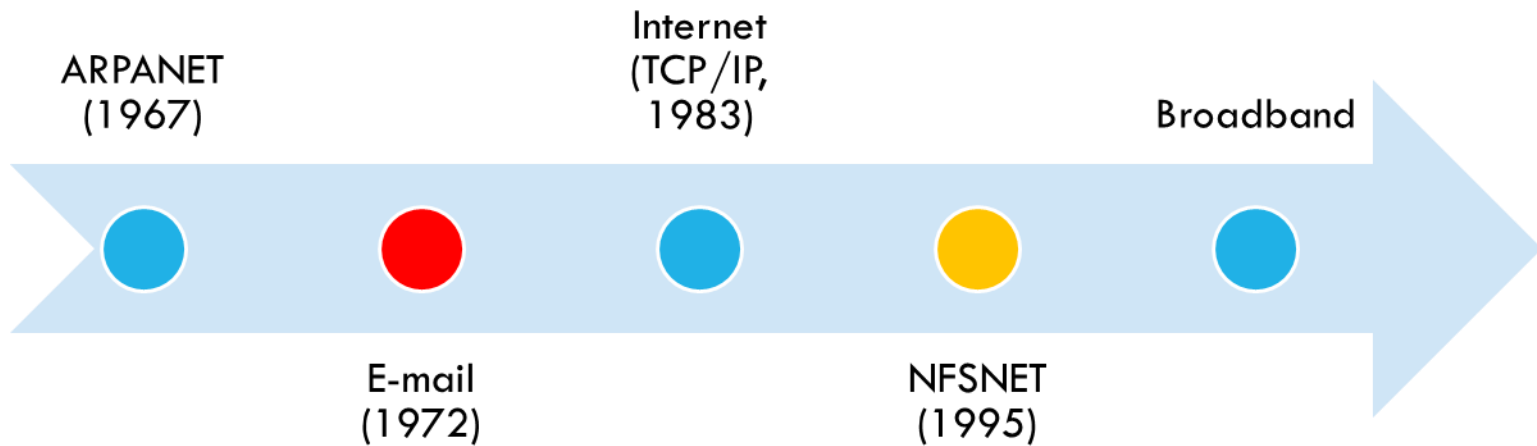
end of 1970s-
Apple computer
and Tandy

1981-IBM PC

MILESTONES IN NETWORKING



MILESTONES IN NETWORKING-CONTINUE



MILESTONE IN INFORMATION STORAGE AND RETRIEVAL



Codex



Newspaper



Graphical
User
Interface

Networked
Hypertext:
world wide
web



Gutenberg's
Press



Hypertext



Single
Computer
Hypertext
systems

Search
Engines



SUMMARIZATION OF MILESTONE IN IT (IT TIMELINE)

So what is the
impact of IT?

IMPACT OF IT

Change of lifestyle

- Change in how we communicate with others (letter/ face to face -> email, internet, FB, twitter)
- Change in how we do business (e-business, e-shopping)
- Change in how we study

INFORMATION TECHNOLOGY ISSUES

Information Technology: *refers to devices used in the creation, storage, manipulation, exchange, and dissemination of data, including text, sound and images.*

Cost of IT devices decrease, while capabilities continue to increase

Use of IT increase->issues because growth of IT

EXAMPLE

Email-Spam: unsolicited (unwanted), bulk, commercial email

Internet: easier to share information but some information that we share might be not legal in other country-> should we stop from posting material?

Unauthorized copies of song, movies and computer program through internet

ETHICS, COMPUTER AND PRIVACY

Database

Networking

Privacy Rules and Law

Online Privacy

PRIVACY PROTECTION AND THE LAW

The use of IT in business requires balancing the needs of those who use the information against the rights and desires of the people whose information may be used

On one hand, information about people is gathered, stored, analyzed and reported because organizations can use it to make better decisions. Organizations need basic information about customers to serve them better.

On the other hand, many object to the data collection policies of government and businesses.

According to U.S. Census data, privacy is a key concern of Internet users and a top reason why nonusers still avoid the Internet.

PRIVACY PROTECTION AND THE LAW

Historical perspective on the right to privacy:

- ▢ U.S. Constitution took effect in 1789
- ▢ Although, the Constitution does not contain the word privacy, the U.S. Supreme Court has ruled that the concept of privacy is protected by a number of amendments in the Bill of Rights.
- ▢ Supreme Court has stated that the American Citizens are protected by the Fourth Amendment when there is a “reasonable expectation of privacy”.

To today, in addition to protection from government intrusion (interference), people need privacy protection from private industry. Few laws provide such protection.

RECENT HISTORY OF PRIVACY PROTECTION

Communications Act of 1934 restricted the government's ability to secretly intercept communications. However, under a 1968 federal statute, law enforcement officers can use wiretapping (monitoring) - the interception of telephone or telegraph communications for purpose of espionage or surveillance - if the first obtain a court order.(investigation)

FOIA - The Freedom of Information Act passed in 1966 and amended in 1974, provides public with the means to gain access to certain government records.

Fair Credit Reporting Act of 1970 regulates the operations of credit-reporting bureaus, including how they collect, store and use credit information.

Privacy Act of 1974 provides certain safeguards for people against invasion of personal privacy by federal agencies. The Central Intelligence Agency (CIA) and law enforcement agencies are excluded from this act; nor does it cover the actions of private industry.

RECENT HISTORY OF PRIVACY PROTECTION

COPA - Children's Online Protection Act was passed by Congress in October 1998.

According to the COPA law, a Web site that caters to children must offer comprehensive privacy policies, notify their parents or guardians about its data collection practices, and receive parent consent before collecting any personal information from children under 13 years of age.

In 2004, the Federal Trade Commission (FTC) accused Bonzi Software Inc. and UMG Recordings Inc. of collecting personal information from children online without their parent's consent, and settled with them for penalties of \$75,000 and \$400, 000

<http://www.ftc.gov/opa/2004/02/bonziumg.htm>

http://en.wikipedia.org/wiki/Bonzi_Buddy

RECENT HISTORY OF PRIVACY PROTECTION

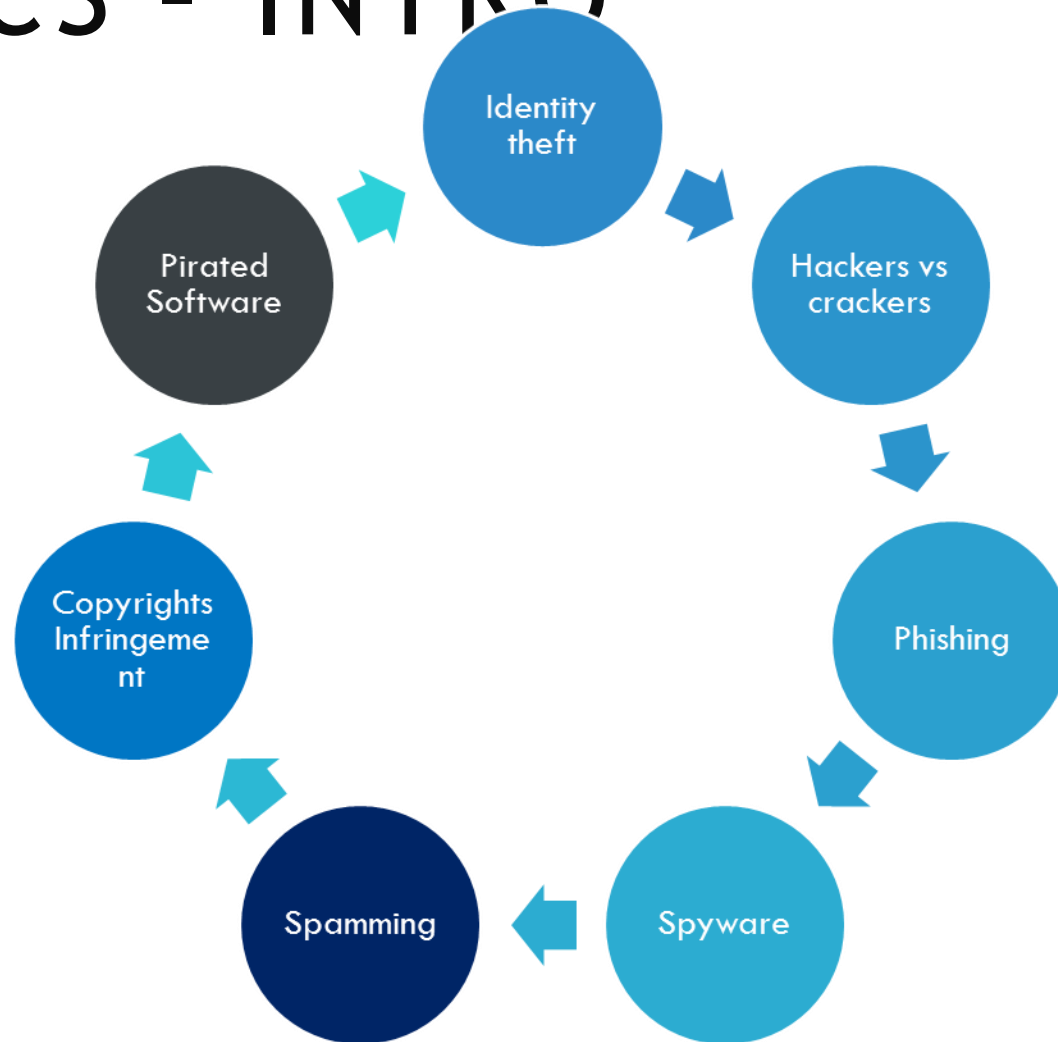
- European Community Directive 95/46/EC of 1998 requires any company that does business within the borders of 15 Western European nations to implement a set of privacy directives on fair and appropriate use of information.
- Gramm-Leach-Bliley Act (1999) - one example of the law that controls opt-out information gathering. The Financial Privacy Rule requires financial institutions to provide each consumer with a privacy notice at the time the consumer relationship is established and annually thereafter.

http://en.wikipedia.org/wiki/Gramm-Leach-Bliley_Act



ISSUES

SOME ISSUES IN IT ETHICS - INTRO

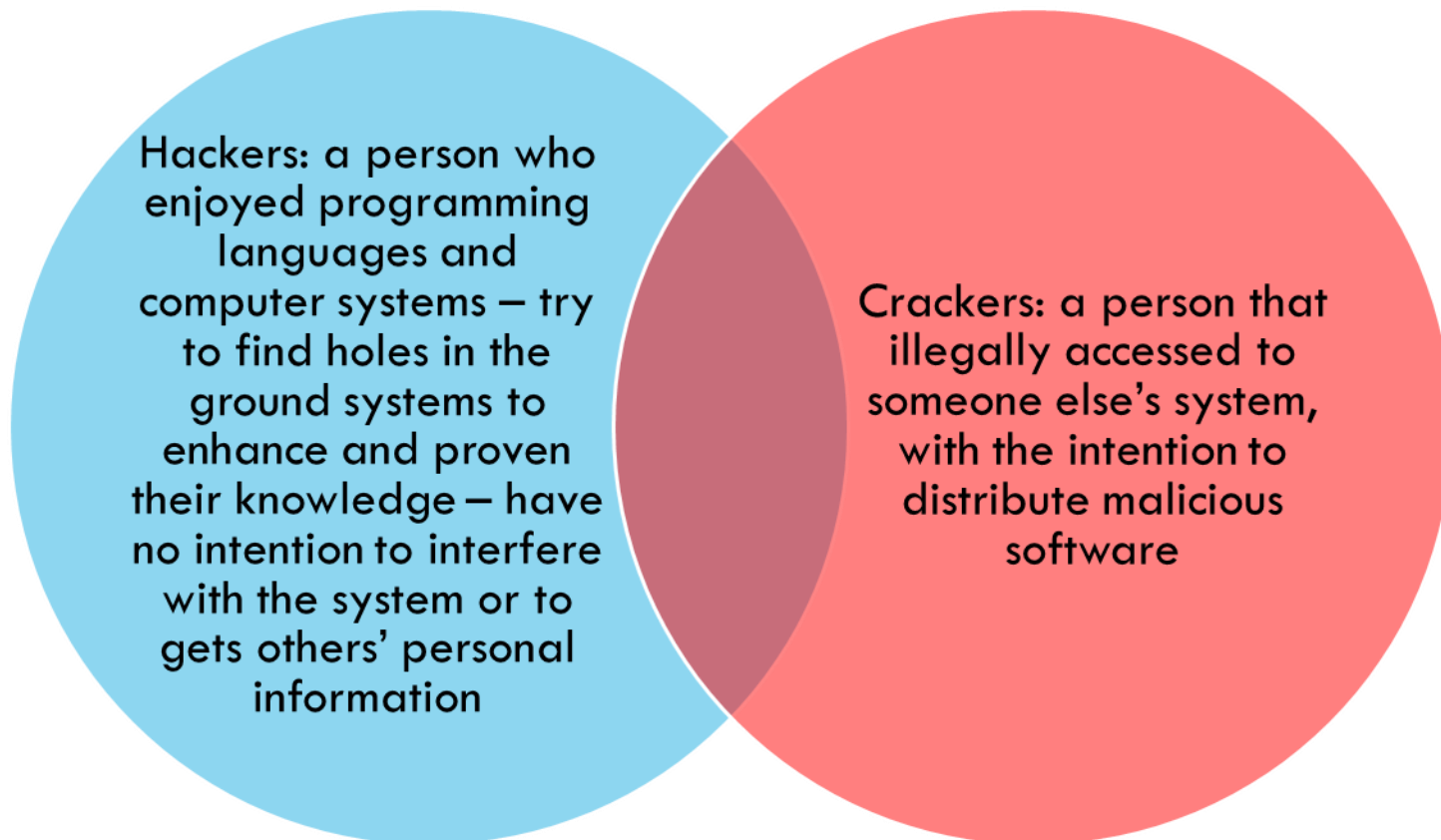


SOME ISSUES IN IT ETHICS - INTRO (CONT)

Identity Theft

- occurs when someone illegally uses your personal information such as name, address or financial information for their own interests
- once cyber thieves access to our information, they can use it
- eg: open credit card account using victim name

SOME ISSUES IN IT ETHICS - INTRO (CONT)



CRACKING OF LARGE DATABASES TO GAIN PERSONAL IDENTITY INFORMATION

Partial list of incidents from 2005:

February 2005, Check Point, keeper of more than 19 million public records, revealed that hackers stole data on more than 147, 000 consumers

March 2005, Reed Elsevier, the parent company of LexisNexis, announced that hackers had compromised its massive database, stealing information on more than 300, 000 people

<http://www.commercialalert.org/issues/culture/privacy/another-data-broker-reports-a-breach>

March 2005, Retail Ventures Inc. reported the theft of credit card data and other personal information of 1.4 million customers from its DSW Store Warehouse stores.

CRACKING OF LARGE DATABASES TO GAIN PERSONAL IDENTITY INFORMATION

March 2005, Bank of America disclosed that it lost credit card account records of 1.2 million federal employees

June 2005, Visa USA and American Express announced that they were terminating their contract with CardSystem Solutions after a hacker accesses as many as 40 million credit card numbers

The number of incidents is alarming.

The lack of the initiative by some companies in informing people whose data was stolen.

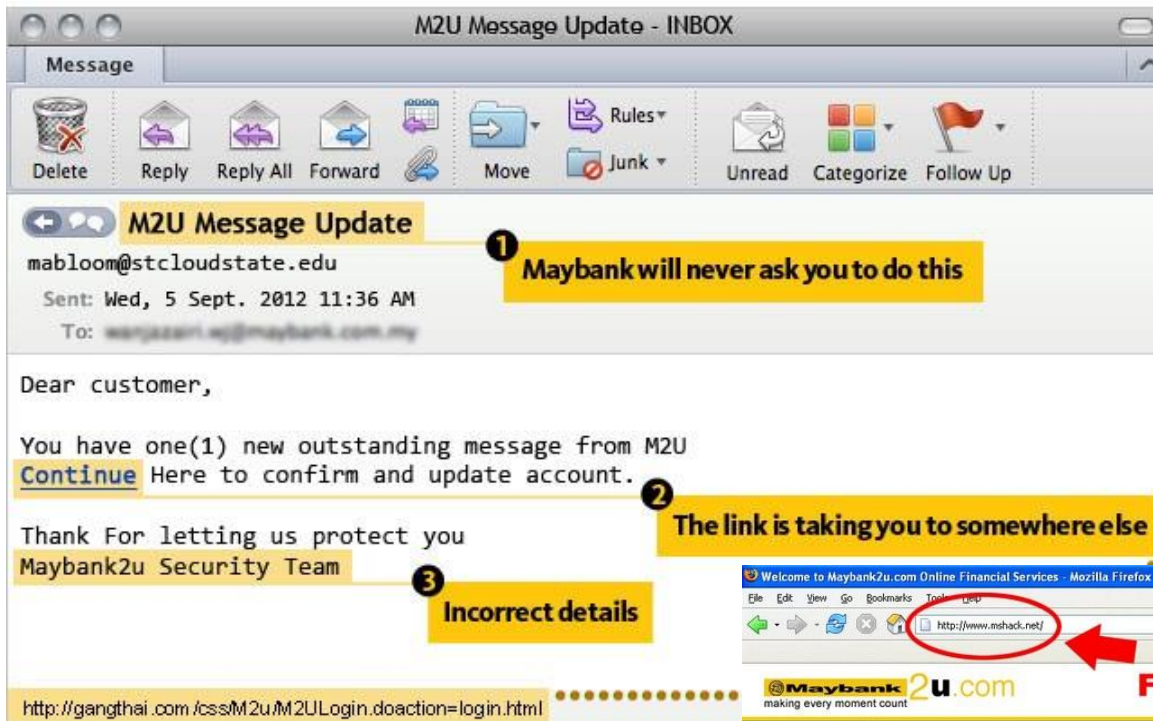
<http://www.computerworld.com/securitytopics/security/story/0,10801,101058,00.html>

SOME ISSUES IN IT ETHICS - INTRO (CONT)

Phishing

- Spoofed (fool) e-mails lead consumers to the fake Web sites
- an attempt to steal personal identity data by tricking users entering the information on a counterfeit (fake) Web site; this data includes credit card numbers, account usernames, passwords, SSN.

PHISHING EXAMPLES



SOME ISSUES IN IT ETHICS - INTRO (CONT)

Spyware

- a term for keystroke-logging software that is downloaded to user's computer without adequate notice, consent, or control for the users. (Spyware is a type of malware that steals personal information from a computer without the user's knowledge)

<http://en.wikipedia.org/wiki/Spyware>

- creates a record of the keystrokes entered on the computer, enabling the capture of account usernames, passwords, credit card numbers, and other sensitive information.

SOME ISSUES IN IT ETHICS - INTRO (CONT)

Spamming

- is the transmission of the same e-mail message to a large number of people. (Spam is any kind of unwanted, unsolicited digital communication that gets sent out in bulk)
- The Controlling the Assault of Non-Solicited Pornography and Marketing (CAN-SPAM) Act went into effect in January 2004.
- Act says that it is legal to spam, provided the message meets a few basic requirements
- Not only has the CAN-SPAM Act failed to slow the flow of junk e-mail, but some believe that it actually has increased the flow of spam, because it legalizes sending of unsolicited e-mail
- <http://www.informationweek.com/showArticle.jhtml?articleID=21401320>

SOME ISSUES IN IT ETHICS - INTRO (CONT)

Copyright Infringement Plagiarism & Pirated Software

- the use of works protected by **copyright** law without permission, **infringing** certain exclusive rights granted to the **copyright** holder, such as the right to reproduce, distribute, display or perform the protected work, or to make derivative works.

SOME ISSUES IN IT ETHICS - INTRO (CONT)

Pirated Software

- duplicated software which is distributed without authorization
- happens when someone makes illegal copies of an original program and sells it



What is your responsibility
and ethics toward IT
specifically on e-access?